Bilderwand – layout candidates

Target 180x120 cm ± 10.0 cm, nominal gap free in [1.5, 6.0] cm, orientation fixed, sizes are measured outer dimensions. Max. gap deviation = largest difference between an actual gap and the layout's nominal gap; notches = parts of the outer border not covered by a frame or a regular gap. Ranked by a combined score (gap uniformity first, border sharpness second, then number of frames).

#	size (cm)	gap	frames	max dev	mean dev	notches	corners	entropy
1	182.0 × 129.0	4	25	3.0	0.60	none	4/4	0.65
2	191.5 × 113.5	1.5	24	2.5	0.73	4 cm	4/4	0.66
3	184.0 × 115.0	2	23	2.2	0.55	4 cm	4/4	0.68
4	177.5 × 113.5	1.5	21	3.0	0.52	none	4/4	0.66
5	191.5 × 121.5	1.5	26	3.5	0.51	none	4/4	0.69
6	182.5 × 130.5	1.5	26	3.5	0.57	none	4/4	0.81
7	179.5 × 120.5	1.5	26	4.0	0.54	none	4/4	0.69
8	176.5 × 125.5	1.5	25	3.0	1.04	4 cm	4/4	0.81
9	174.5 × 123.5	2.5	24	3.3	0.56	none	4/4	0.74
10	173.5 × 124.5	2.5	24	3.0	0.71	4 cm	4/4	0.81
11	169.0 × 116.5	2	24	2.5	0.69	8 cm	4/4	0.76
12	178.5 × 111.5	1.5	24	3.8	0.55	none	4/4	0.76
13	170.0 × 123.0	2	24	2.8	0.67	8 cm	4/4	0.82
14	174.5 × 120.5	1.5	24	3.0	0.44	7 cm	4/4	0.80
15	189.0 × 123.0	2	24	4.2	0.50	none	4/4	0.62
16	170.5 × 126.5	2.5	23	3.5	0.77	none	4/4	0.72

Inventory

	outer w × h (cm)		have
	20.5×15.5	L	9
	15.5×20.5	P	2
	32.5×42.5	P	2
	42.5×32.5	L	1
	32.5×23.5	L	5
	23.5×32.5	P	1
	25.5×25.5		5
	42.5×52.5	P	1
	52.5×42.5	L	2

Search parameters: target = 180x120, tol = 10.0, gap_min = 1.5, gap_max = 6.0, gap_mode = free, delta = 0.5, slack_rule = even, cost_shape = huber, min_frames = 10, frame_bonus = 1.5, runs = 256, iters = 20 min, seed = 0, large_area = 2000.0, large_count = 2-3, large_apart = True, large_min_dist = 70.0

Layout 1 182.0 × 129.0 cm gap 4 cm 25 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

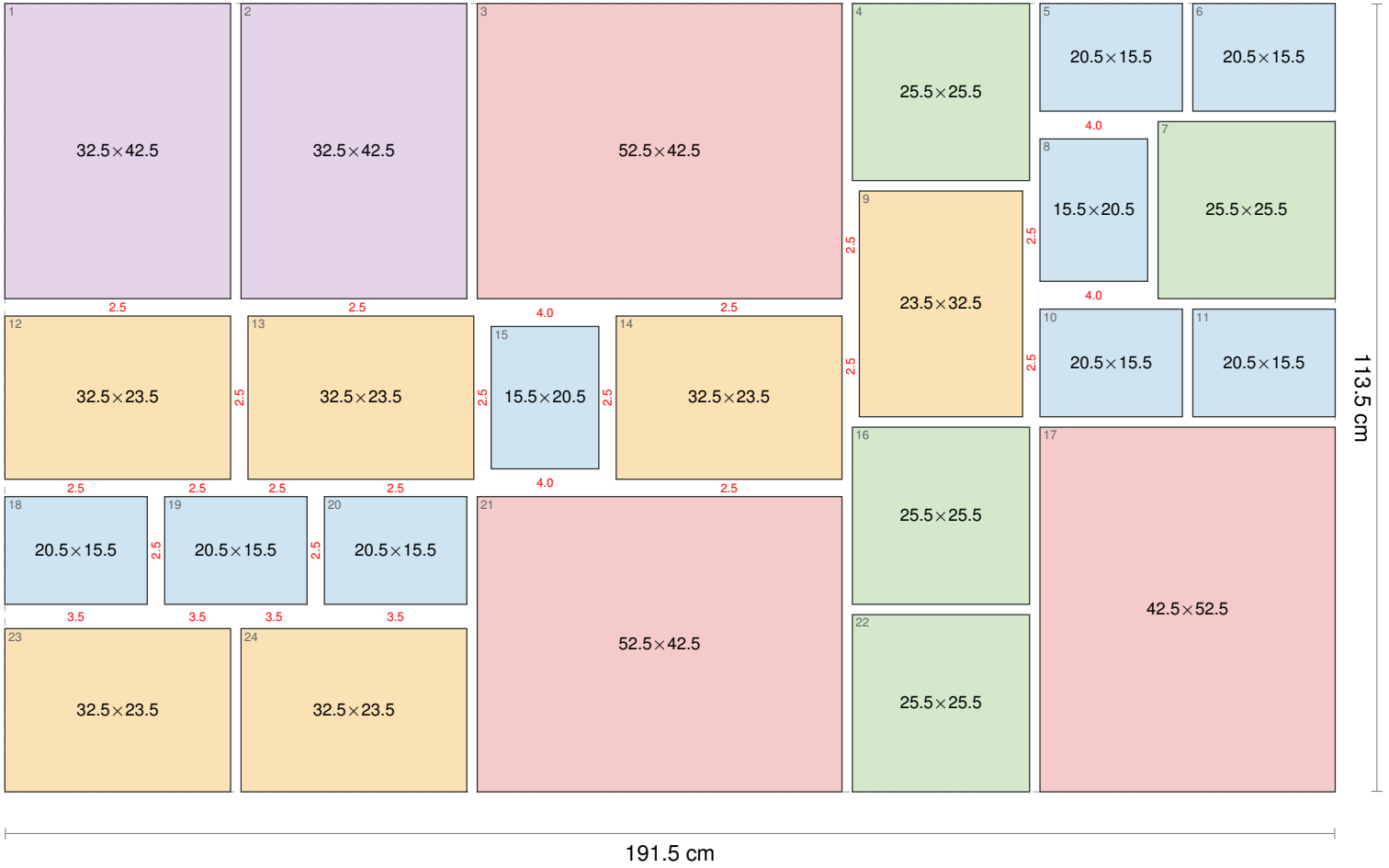
Outer size	182.0 × 129.0 cm
Nominal gap	4 cm
Actual gaps	4.0 – 7.0 cm
Max. deviation	3.0 cm
Mean deviation	0.60 cm
Border notches	none
Corners filled	4 / 4
Frames used	25 of 28
Large frames	2, not touching, min. dist. 132 cm
Entropy	0.65
mix / variety	0.83 / 0.54
stagger / lines	0.48 / 0.75 (2 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	2	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	4	5
	23.5×32.5	P	1	1
	25.5×25.5		4	5
	42.5×52.5	P	0	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	52.5 × 42.5	0.0	0.0
2	32.5 × 23.5	56.5	0.0
3	32.5 × 23.5	93.0	0.0
4	15.5 × 20.5	129.5	0.0
5	32.5 × 23.5	149.5	0.0
6	32.5 × 42.5	56.5	27.5
7	32.5 × 42.5	93.0	27.5
8	52.5 × 42.5	129.5	27.5
9	32.5 × 23.5	0.0	46.5
10	15.5 × 20.5	37.0	48.0
11	25.5 × 25.5	0.0	74.0
12	25.5 × 25.5	30.3	74.0
13	20.5 × 15.5	136.2	74.0
14	20.5 × 15.5	161.5	74.0
15	23.5 × 32.5	60.6	75.2
16	42.5 × 32.5	88.9	75.2
17	20.5 × 15.5	136.2	93.5
18	20.5 × 15.5	161.5	93.5
19	25.5 × 25.5	0.0	103.5
20	25.5 × 25.5	30.3	103.5
21	20.5 × 15.5	60.6	113.5
22	20.5 × 15.5	85.8	113.5
23	20.5 × 15.5	111.0	113.5
24	20.5 × 15.5	136.3	113.5
25	20.5 × 15.5	161.5	113.5

Layout 2 191.5 × 113.5 cm gap 1.5 cm 24 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

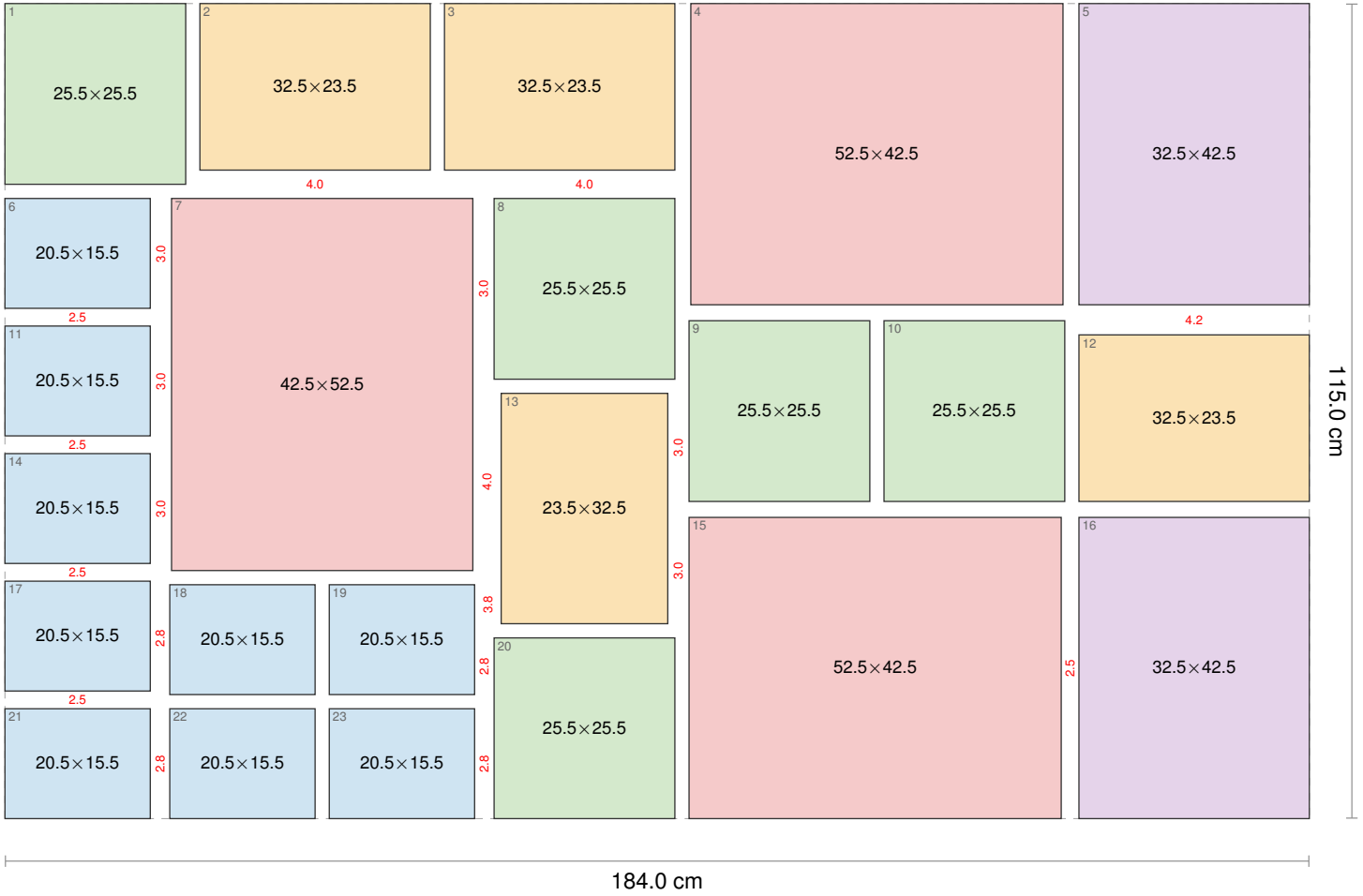
Outer size	191.5 × 113.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 4.0 cm
Max. deviation	2.5 cm
Mean deviation	0.73 cm
Border notches	left 4 cm
Corners filled	4 / 4
Frames used	24 of 28
Large frames	3, not touching, min. dist. 71 cm
Entropy	0.66
mix / variety	0.85 / 0.80
stagger / lines	0.47 / 0.50 (3 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	7	9
	15.5×20.5	P	2	2
	32.5×42.5	P	2	2
	42.5×32.5	L	0	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		4	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	32.5 × 42.5	0.0	0.0
2	32.5 × 42.5	34.0	0.0
3	52.5 × 42.5	68.0	0.0
4	25.5 × 25.5	122.0	0.0
5	20.5 × 15.5	149.0	0.0
6	20.5 × 15.5	171.0	0.0
7	25.5 × 25.5	166.0	17.0
8	15.5 × 20.5	149.0	19.5
9	23.5 × 32.5	123.0	27.0
10	20.5 × 15.5	149.0	44.0
11	20.5 × 15.5	171.0	44.0
12	32.5 × 23.5	0.0	45.0
13	32.5 × 23.5	35.0	45.0
14	32.5 × 23.5	88.0	45.0
15	15.5 × 20.5	70.0	46.5
16	25.5 × 25.5	122.0	61.0
17	42.5 × 52.5	149.0	61.0
18	20.5 × 15.5	0.0	71.0
19	20.5 × 15.5	23.0	71.0
20	20.5 × 15.5	46.0	71.0
21	52.5 × 42.5	68.0	71.0
22	25.5 × 25.5	122.0	88.0
23	32.5 × 23.5	0.0	90.0
24	32.5 × 23.5	34.0	90.0

Layout 3 184.0 × 115.0 cm gap 2 cm 23 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

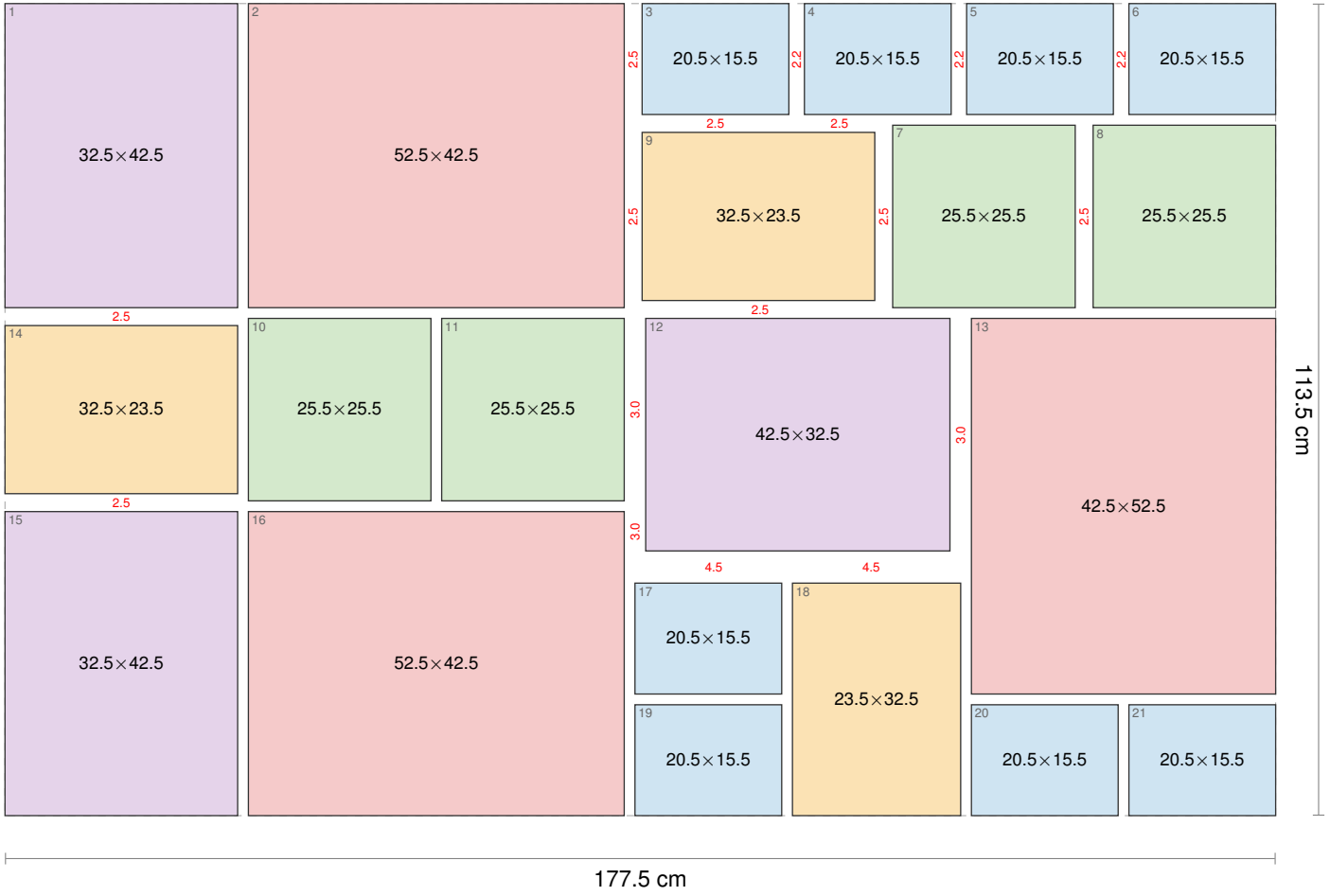
Outer size	184.0 × 115.0 cm
Nominal gap	2 cm
Actual gaps	2.0 – 4.2 cm
Max. deviation	2.2 cm
Mean deviation	0.55 cm
Border notches	right 4 cm
Corners filled	4 / 4
Frames used	23 of 28
Large frames	3, not touching, min. dist. 73 cm
Entropy	0.68
mix / variety	0.76 / 0.66
stagger / lines	0.54 / 0.75 (2 through)

outer w × h (cm)		used	have
20.5×15.5	L	9	9
15.5×20.5	P	0	2
32.5×42.5	P	2	2
42.5×32.5	L	0	1
32.5×23.5	L	3	5
23.5×32.5	P	1	1
25.5×25.5		5	5
42.5×52.5	P	1	1
52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	25.5 × 25.5	0.0	0.0
2	32.5 × 23.5	27.5	0.0
3	32.5 × 23.5	62.0	0.0
4	52.5 × 42.5	96.8	0.0
5	32.5 × 42.5	151.5	0.0
6	20.5 × 15.5	0.0	27.5
7	42.5 × 52.5	23.5	27.5
8	25.5 × 25.5	69.0	27.5
9	25.5 × 25.5	96.5	44.8
10	25.5 × 25.5	124.0	44.8
11	20.5 × 15.5	0.0	45.5
12	32.5 × 23.5	151.5	46.8
13	23.5 × 32.5	70.0	55.0
14	20.5 × 15.5	0.0	63.5
15	52.5 × 42.5	96.5	72.5
16	32.5 × 42.5	151.5	72.5
17	20.5 × 15.5	0.0	81.5
18	20.5 × 15.5	23.2	82.0
19	20.5 × 15.5	45.8	82.0
20	25.5 × 25.5	69.0	89.5
21	20.5 × 15.5	0.0	99.5
22	20.5 × 15.5	23.2	99.5
23	20.5 × 15.5	45.8	99.5

Layout 4 177.5 × 113.5 cm gap 1.5 cm 21 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

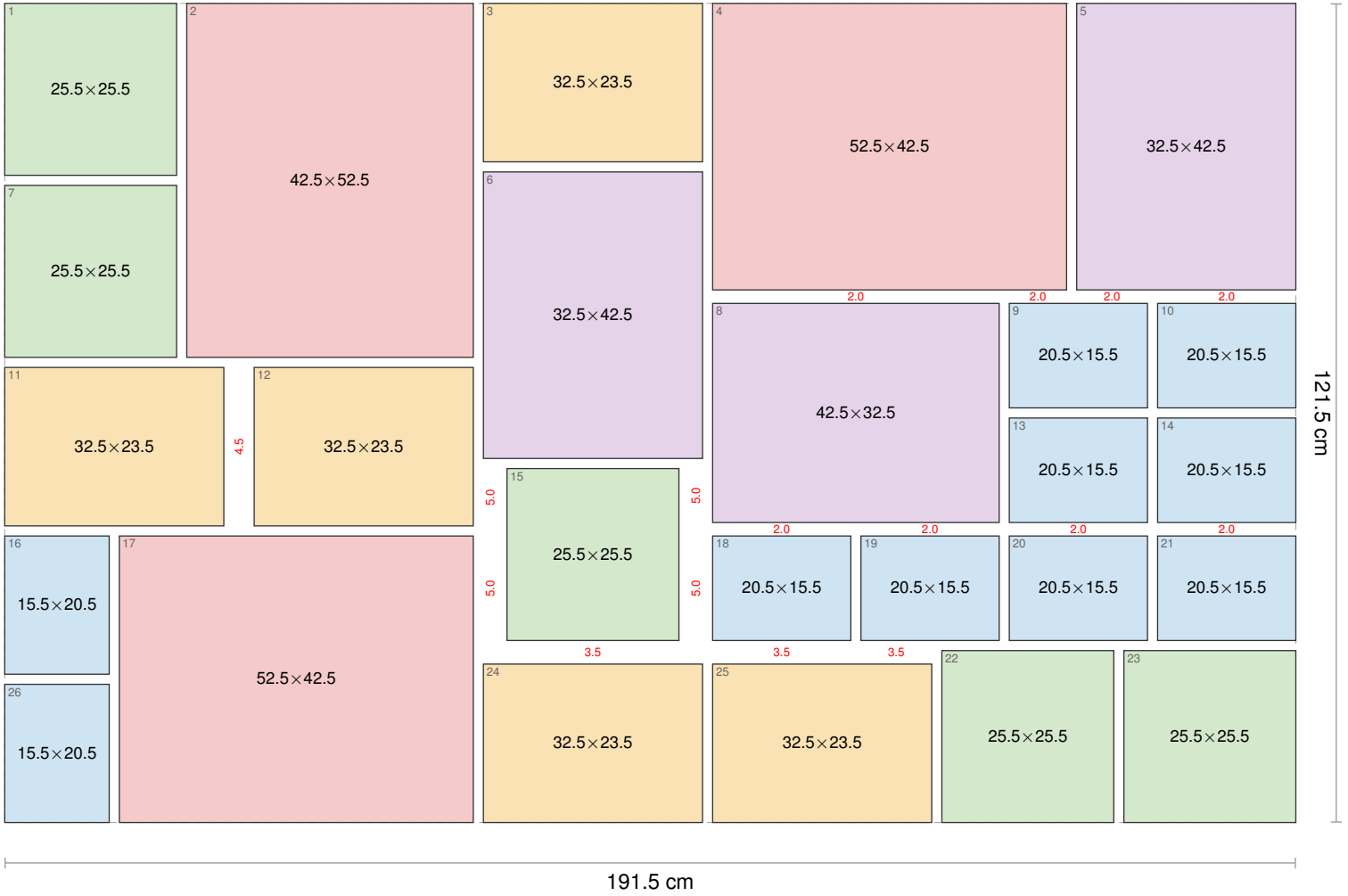
Outer size	177.5 × 113.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 4.5 cm
Max. deviation	3.0 cm
Mean deviation	0.52 cm
Border notches	none
Corners filled	4 / 4
Frames used	21 of 28
Large frames	3, not touching, min. dist. 71 cm
Entropy	0.66
mix / variety	0.81 / 0.81
stagger / lines	0.50 / 0.50 (3 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	8	9
	15.5×20.5	P	0	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	2	5
	23.5×32.5	P	1	1
	25.5×25.5		4	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	32.5 × 42.5	0.0	0.0
2	52.5 × 42.5	34.0	0.0
3	20.5 × 15.5	89.0	0.0
4	20.5 × 15.5	111.7	0.0
5	20.5 × 15.5	134.3	0.0
6	20.5 × 15.5	157.0	0.0
7	25.5 × 25.5	124.0	17.0
8	25.5 × 25.5	152.0	17.0
9	32.5 × 23.5	89.0	18.0
10	25.5 × 25.5	34.0	44.0
11	25.5 × 25.5	61.0	44.0
12	42.5 × 32.5	89.5	44.0
13	42.5 × 52.5	135.0	44.0
14	32.5 × 23.5	0.0	45.0
15	32.5 × 42.5	0.0	71.0
16	52.5 × 42.5	34.0	71.0
17	20.5 × 15.5	88.0	81.0
18	23.5 × 32.5	110.0	81.0
19	20.5 × 15.5	88.0	98.0
20	20.5 × 15.5	135.0	98.0
21	20.5 × 15.5	157.0	98.0

Layout 5 191.5 × 121.5 cm gap 1.5 cm 26 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

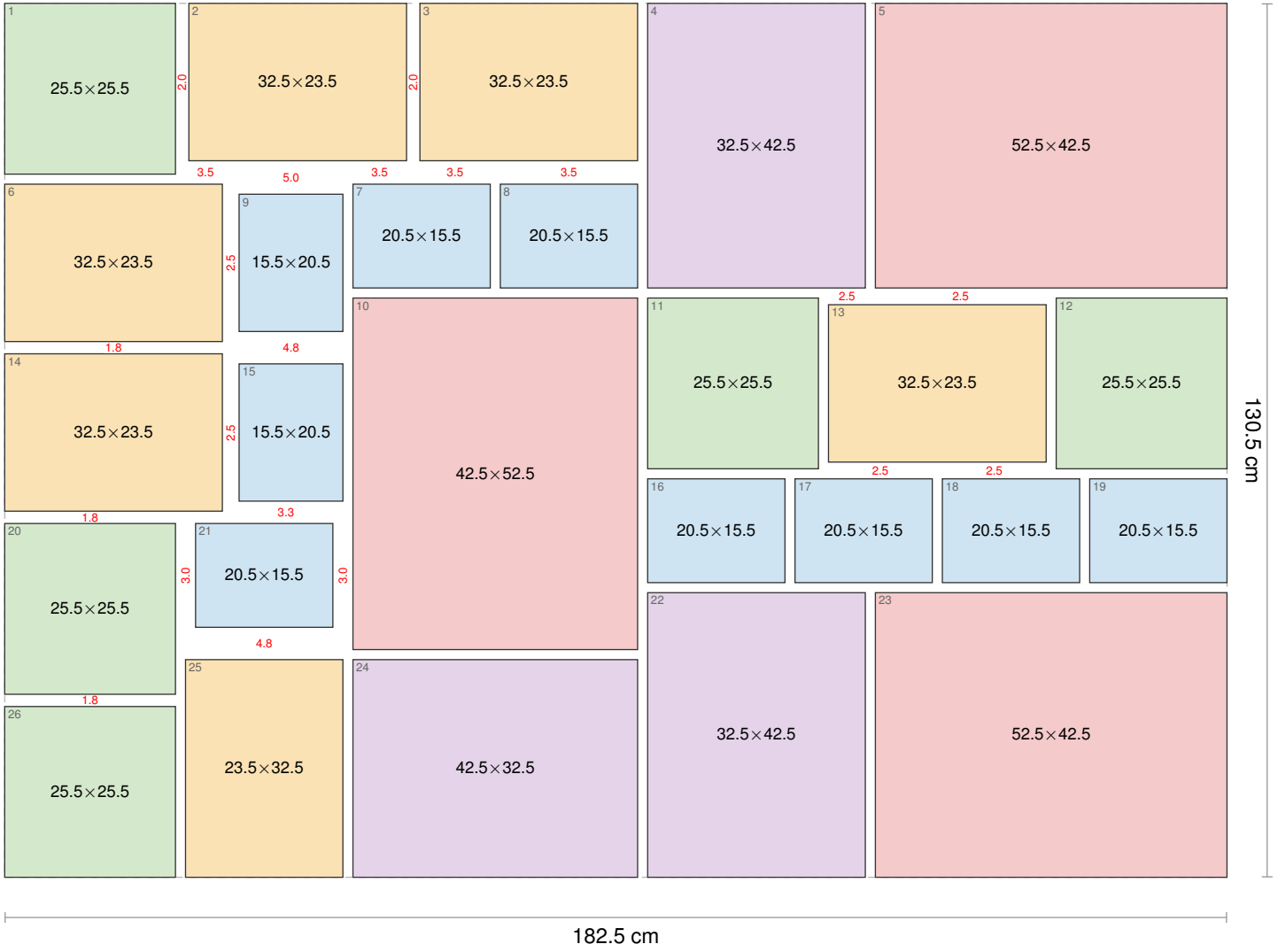
Outer size	191.5 × 121.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 5.0 cm
Max. deviation	3.5 cm
Mean deviation	0.51 cm
Border notches	none
Corners filled	4 / 4
Frames used	26 of 28
Large frames	3, not touching, min. dist. 74 cm
Entropy	0.69
mix / variety	0.84 / 0.71
stagger / lines	0.46 / 0.75 (2 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	8	9
	15.5×20.5	P	2	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	0	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	25.5 × 25.5	0.0	0.0
2	42.5 × 52.5	27.0	0.0
3	32.5 × 23.5	71.0	0.0
4	52.5 × 42.5	105.0	0.0
5	32.5 × 42.5	159.0	0.0
6	32.5 × 42.5	71.0	25.0
7	25.5 × 25.5	0.0	27.0
8	42.5 × 32.5	105.0	44.5
9	20.5 × 15.5	149.0	44.5
10	20.5 × 15.5	171.0	44.5
11	32.5 × 23.5	0.0	54.0
12	32.5 × 23.5	37.0	54.0
13	20.5 × 15.5	149.0	61.5
14	20.5 × 15.5	171.0	61.5
15	25.5 × 25.5	74.5	69.0
16	15.5 × 20.5	0.0	79.0
17	52.5 × 42.5	17.0	79.0
18	20.5 × 15.5	105.0	79.0
19	20.5 × 15.5	127.0	79.0
20	20.5 × 15.5	149.0	79.0
21	20.5 × 15.5	171.0	79.0
22	25.5 × 25.5	139.0	96.0
23	25.5 × 25.5	166.0	96.0
24	32.5 × 23.5	71.0	98.0
25	32.5 × 23.5	105.0	98.0
26	15.5 × 20.5	0.0	101.0

Layout 6 182.5 × 130.5 cm gap 1.5 cm 26 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

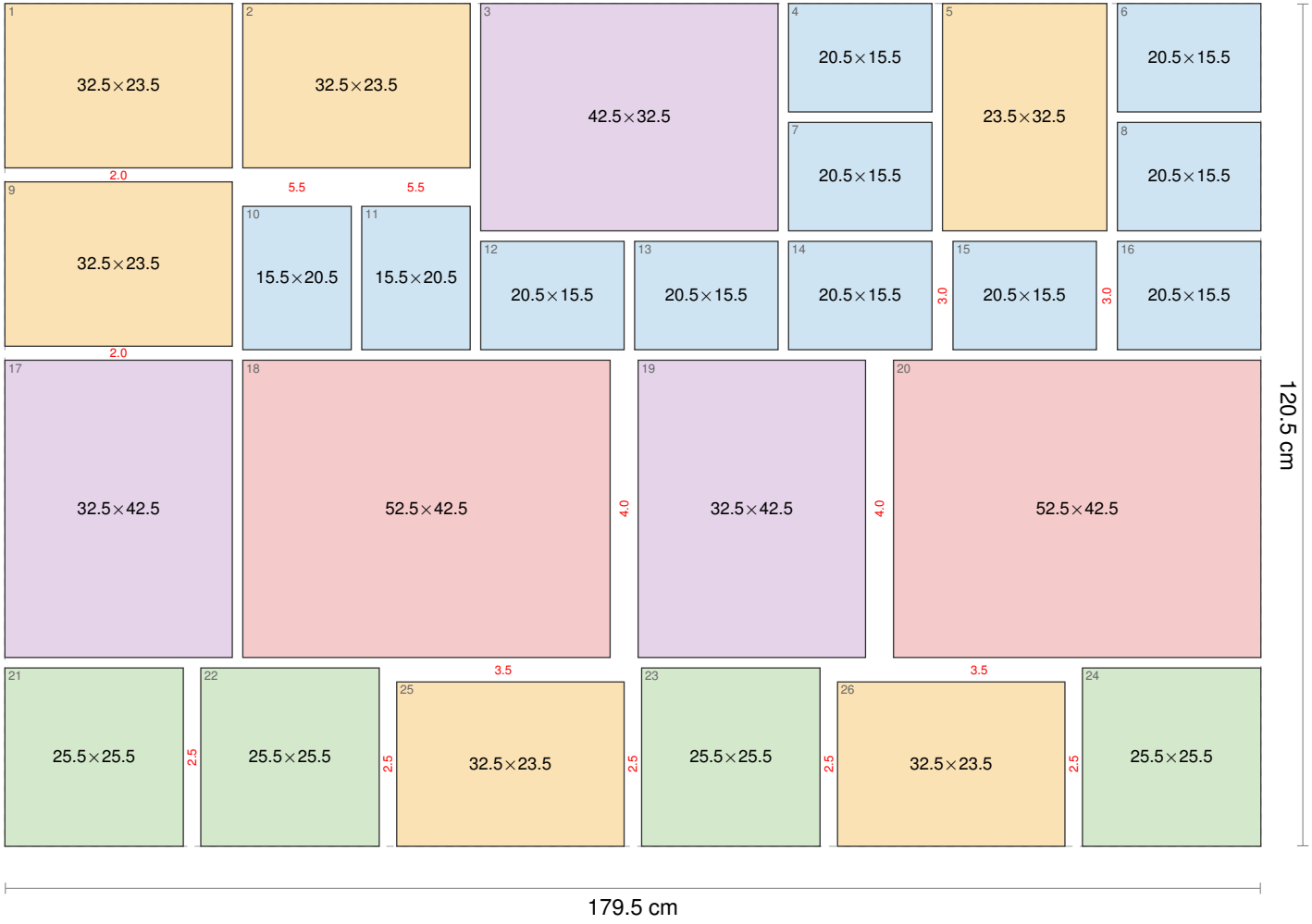
Outer size	182.5 × 130.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 5.0 cm
Max. deviation	3.5 cm
Mean deviation	0.57 cm
Border notches	none
Corners filled	4 / 4
Frames used	26 of 28
Large frames	3, not touching, min. dist. 88 cm
Entropy	0.81
mix / variety	0.89 / 0.84
stagger / lines	0.53 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	7	9
	15.5×20.5	P	2	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	25.5 × 25.5	0.0	0.0
2	32.5 × 23.5	27.5	0.0
3	32.5 × 23.5	62.0	0.0
4	32.5 × 42.5	96.0	0.0
5	52.5 × 42.5	130.0	0.0
6	32.5 × 23.5	0.0	27.0
7	20.5 × 15.5	52.0	27.0
8	20.5 × 15.5	74.0	27.0
9	15.5 × 20.5	35.0	28.5
10	42.5 × 52.5	52.0	44.0
11	25.5 × 25.5	96.0	44.0
12	25.5 × 25.5	157.0	44.0
13	32.5 × 23.5	123.0	45.0
14	32.5 × 23.5	0.0	52.3
15	15.5 × 20.5	35.0	53.8
16	20.5 × 15.5	96.0	71.0
17	20.5 × 15.5	118.0	71.0
18	20.5 × 15.5	140.0	71.0
19	20.5 × 15.5	162.0	71.0
20	25.5 × 25.5	0.0	77.7
21	20.5 × 15.5	28.5	77.7
22	32.5 × 42.5	96.0	88.0
23	52.5 × 42.5	130.0	88.0
24	42.5 × 32.5	52.0	98.0
25	23.5 × 32.5	27.0	98.0
26	25.5 × 25.5	0.0	105.0

Layout 7 179.5 × 120.5 cm gap 1.5 cm 26 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

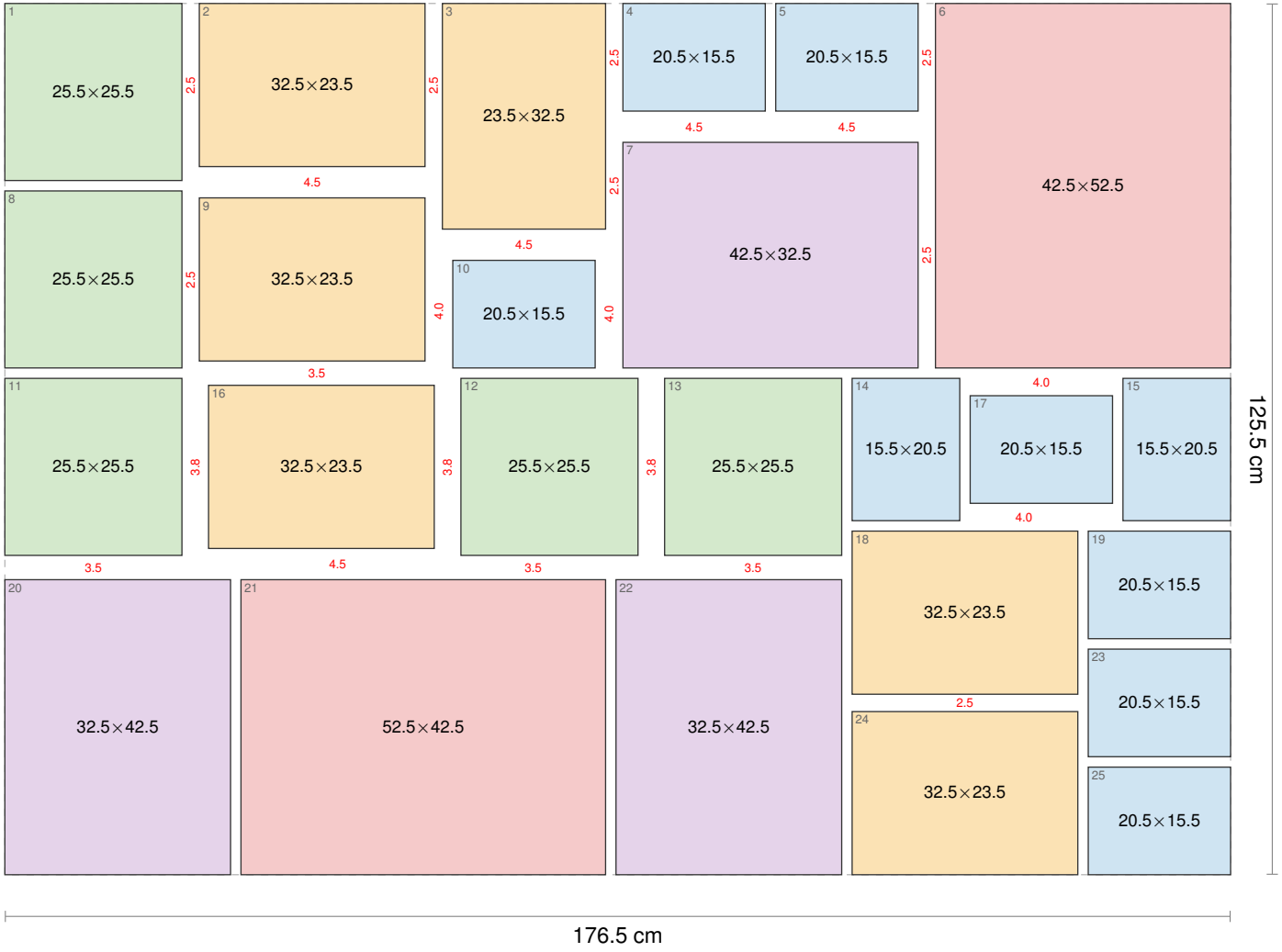
Outer size	179.5 × 120.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 5.5 cm
Max. deviation	4.0 cm
Mean deviation	0.54 cm
Border notches	none
Corners filled	4 / 4
Frames used	26 of 28
Large frames	2, not touching, min. dist. 93 cm
Entropy	0.69
mix / variety	0.83 / 0.75
stagger / lines	0.44 / 0.75 (2 through)

outer w × h (cm)		used	have
20.5×15.5	L	9	9
15.5×20.5	P	2	2
32.5×42.5	P	2	2
42.5×32.5	L	1	1
32.5×23.5	L	5	5
23.5×32.5	P	1	1
25.5×25.5		4	5
42.5×52.5	P	0	1
52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	32.5 × 23.5	0.0	0.0
2	32.5 × 23.5	34.0	0.0
3	42.5 × 32.5	68.0	0.0
4	20.5 × 15.5	112.0	0.0
5	23.5 × 32.5	134.0	0.0
6	20.5 × 15.5	159.0	0.0
7	20.5 × 15.5	112.0	17.0
8	20.5 × 15.5	159.0	17.0
9	32.5 × 23.5	0.0	25.5
10	15.5 × 20.5	34.0	29.0
11	15.5 × 20.5	51.0	29.0
12	20.5 × 15.5	68.0	34.0
13	20.5 × 15.5	90.0	34.0
14	20.5 × 15.5	112.0	34.0
15	20.5 × 15.5	135.5	34.0
16	20.5 × 15.5	159.0	34.0
17	32.5 × 42.5	0.0	51.0
18	52.5 × 42.5	34.0	51.0
19	32.5 × 42.5	90.5	51.0
20	52.5 × 42.5	127.0	51.0
21	25.5 × 25.5	0.0	95.0
22	25.5 × 25.5	28.0	95.0
23	25.5 × 25.5	91.0	95.0
24	25.5 × 25.5	154.0	95.0
25	32.5 × 23.5	56.0	97.0
26	32.5 × 23.5	119.0	97.0

Layout 8 176.5 × 125.5 cm gap 1.5 cm 25 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

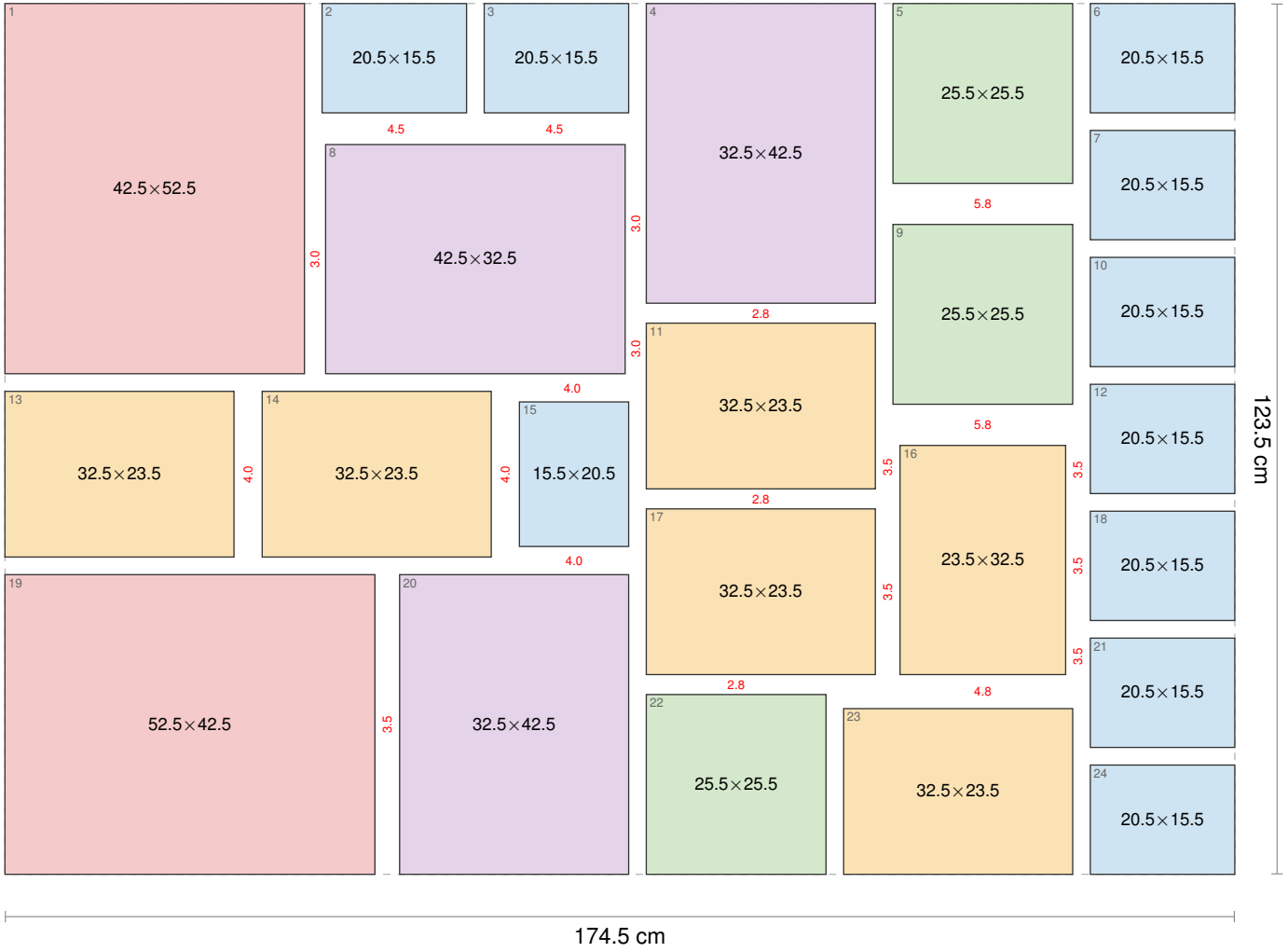
Outer size	176.5 × 125.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 4.5 cm
Max. deviation	3.0 cm
Mean deviation	1.04 cm
Border notches	left 4 cm
Corners filled	4 / 4
Frames used	25 of 28
Large frames	2, not touching, min. dist. 123 cm
Entropy	0.81
mix / variety	0.87 / 0.80
stagger / lines	0.57 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	7	9
	15.5×20.5	P	2	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	1	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	25.5 × 25.5	0.0	0.0
2	32.5 × 23.5	28.0	0.0
3	23.5 × 32.5	63.0	0.0
4	20.5 × 15.5	89.0	0.0
5	20.5 × 15.5	111.0	0.0
6	42.5 × 52.5	134.0	0.0
7	42.5 × 32.5	89.0	20.0
8	25.5 × 25.5	0.0	27.0
9	32.5 × 23.5	28.0	28.0
10	20.5 × 15.5	64.5	37.0
11	25.5 × 25.5	0.0	54.0
12	25.5 × 25.5	65.7	54.0
13	25.5 × 25.5	95.0	54.0
14	15.5 × 20.5	122.0	54.0
15	15.5 × 20.5	161.0	54.0
16	32.5 × 23.5	29.3	55.0
17	20.5 × 15.5	139.0	56.5
18	32.5 × 23.5	122.0	76.0
19	20.5 × 15.5	156.0	76.0
20	32.5 × 42.5	0.0	83.0
21	52.5 × 42.5	34.0	83.0
22	32.5 × 42.5	88.0	83.0
23	20.5 × 15.5	156.0	93.0
24	32.5 × 23.5	122.0	102.0
25	20.5 × 15.5	156.0	110.0

Layout 9 174.5 × 123.5 cm gap 2.5 cm 24 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

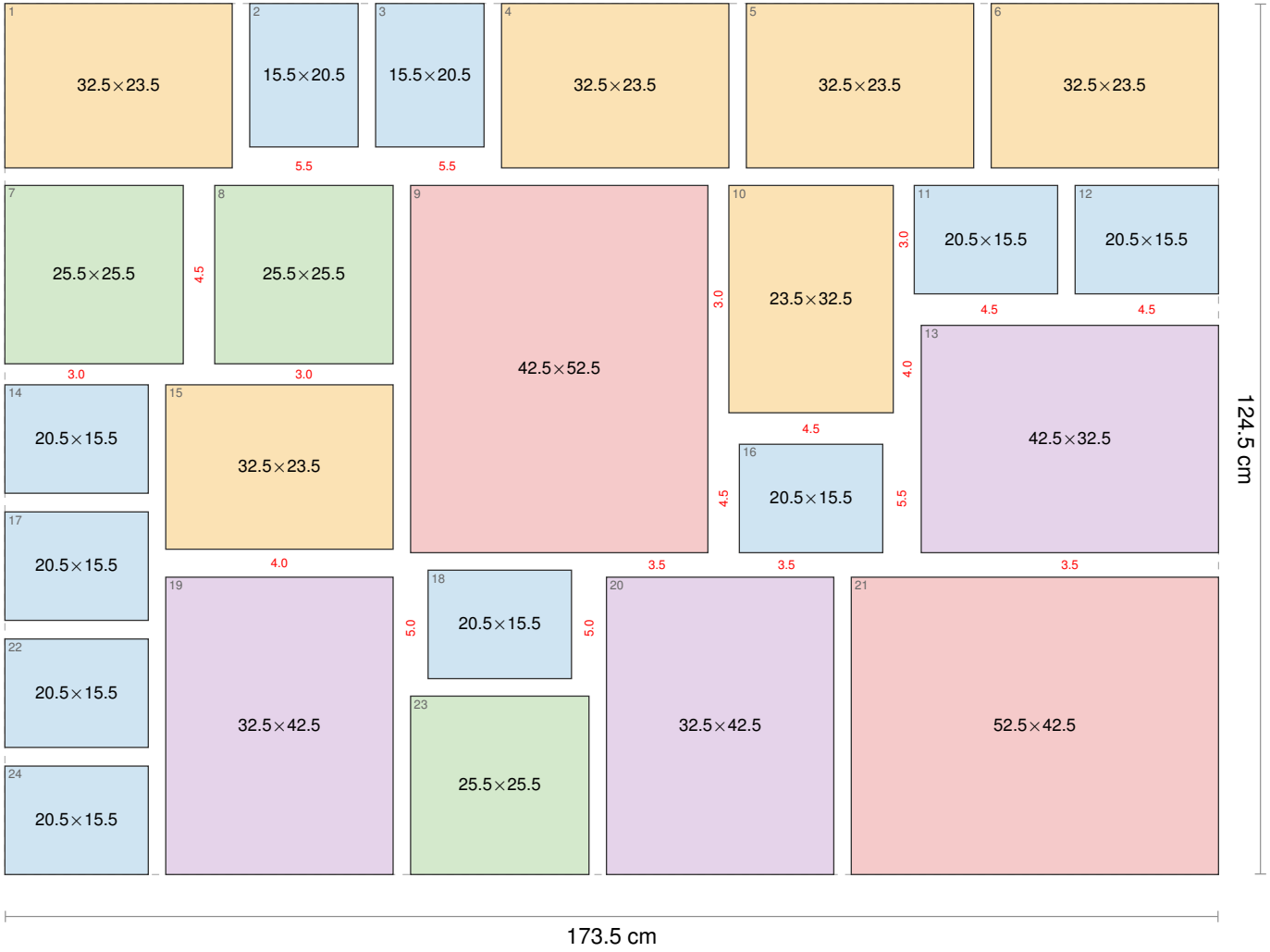
Outer size	174.5 × 123.5 cm
Nominal gap	2.5 cm
Actual gaps	2.5 – 5.8 cm
Max. deviation	3.3 cm
Mean deviation	0.56 cm
Border notches	none
Corners filled	4 / 4
Frames used	24 of 28
Large frames	2, not touching, min. dist. 76 cm
Entropy	0.74
mix / variety	0.83 / 0.80
stagger / lines	0.58 / 0.75 (2 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	1	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		3	5
	42.5×52.5	P	1	1
	52.5×42.5	L	1	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	42.5 × 52.5	0.0	0.0
2	20.5 × 15.5	45.0	0.0
3	20.5 × 15.5	68.0	0.0
4	32.5 × 42.5	91.0	0.0
5	25.5 × 25.5	126.0	0.0
6	20.5 × 15.5	154.0	0.0
7	20.5 × 15.5	154.0	18.0
8	42.5 × 32.5	45.5	20.0
9	25.5 × 25.5	126.0	31.3
10	20.5 × 15.5	154.0	36.0
11	32.5 × 23.5	91.0	45.3
12	20.5 × 15.5	154.0	54.0
13	32.5 × 23.5	0.0	55.0
14	32.5 × 23.5	36.5	55.0
15	15.5 × 20.5	73.0	56.5
16	23.5 × 32.5	127.0	62.7
17	32.5 × 23.5	91.0	71.7
18	20.5 × 15.5	154.0	72.0
19	52.5 × 42.5	0.0	81.0
20	32.5 × 42.5	56.0	81.0
21	20.5 × 15.5	154.0	90.0
22	25.5 × 25.5	91.0	98.0
23	32.5 × 23.5	119.0	100.0
24	20.5 × 15.5	154.0	108.0

Layout 10 173.5 × 124.5 cm gap 2.5 cm 24 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

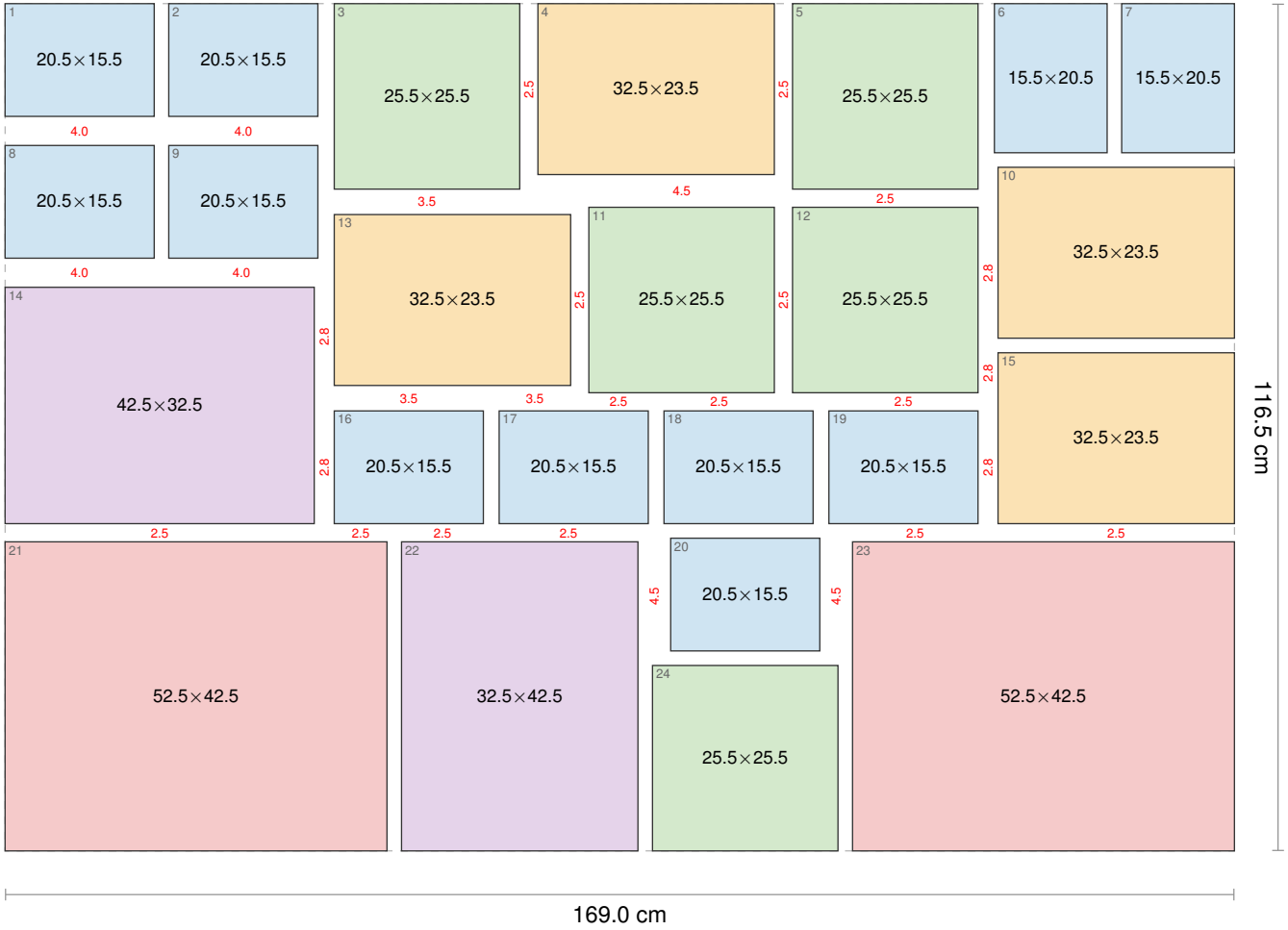
Outer size	173.5 × 124.5 cm
Nominal gap	2.5 cm
Actual gaps	2.5 – 5.5 cm
Max. deviation	3.0 cm
Mean deviation	0.71 cm
Border notches	right 4 cm
Corners filled	4 / 4
Frames used	24 of 28
Large frames	2, not touching, min. dist. 85 cm
Entropy	0.81
mix / variety	0.86 / 0.79
stagger / lines	0.57 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	8	9
	15.5×20.5	P	2	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		3	5
	42.5×52.5	P	1	1
	52.5×42.5	L	1	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	32.5 × 23.5	0.0	0.0
2	15.5 × 20.5	35.0	0.0
3	15.5 × 20.5	53.0	0.0
4	32.5 × 23.5	71.0	0.0
5	32.5 × 23.5	106.0	0.0
6	32.5 × 23.5	141.0	0.0
7	25.5 × 25.5	0.0	26.0
8	25.5 × 25.5	30.0	26.0
9	42.5 × 52.5	58.0	26.0
10	23.5 × 32.5	103.5	26.0
11	20.5 × 15.5	130.0	26.0
12	20.5 × 15.5	153.0	26.0
13	42.5 × 32.5	131.0	46.0
14	20.5 × 15.5	0.0	54.5
15	32.5 × 23.5	23.0	54.5
16	20.5 × 15.5	105.0	63.0
17	20.5 × 15.5	0.0	72.7
18	20.5 × 15.5	60.5	81.0
19	32.5 × 42.5	23.0	82.0
20	32.5 × 42.5	86.0	82.0
21	52.5 × 42.5	121.0	82.0
22	20.5 × 15.5	0.0	90.8
23	25.5 × 25.5	58.0	99.0
24	20.5 × 15.5	0.0	109.0

Layout 11 169.0 × 116.5 cm gap 2 cm 24 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

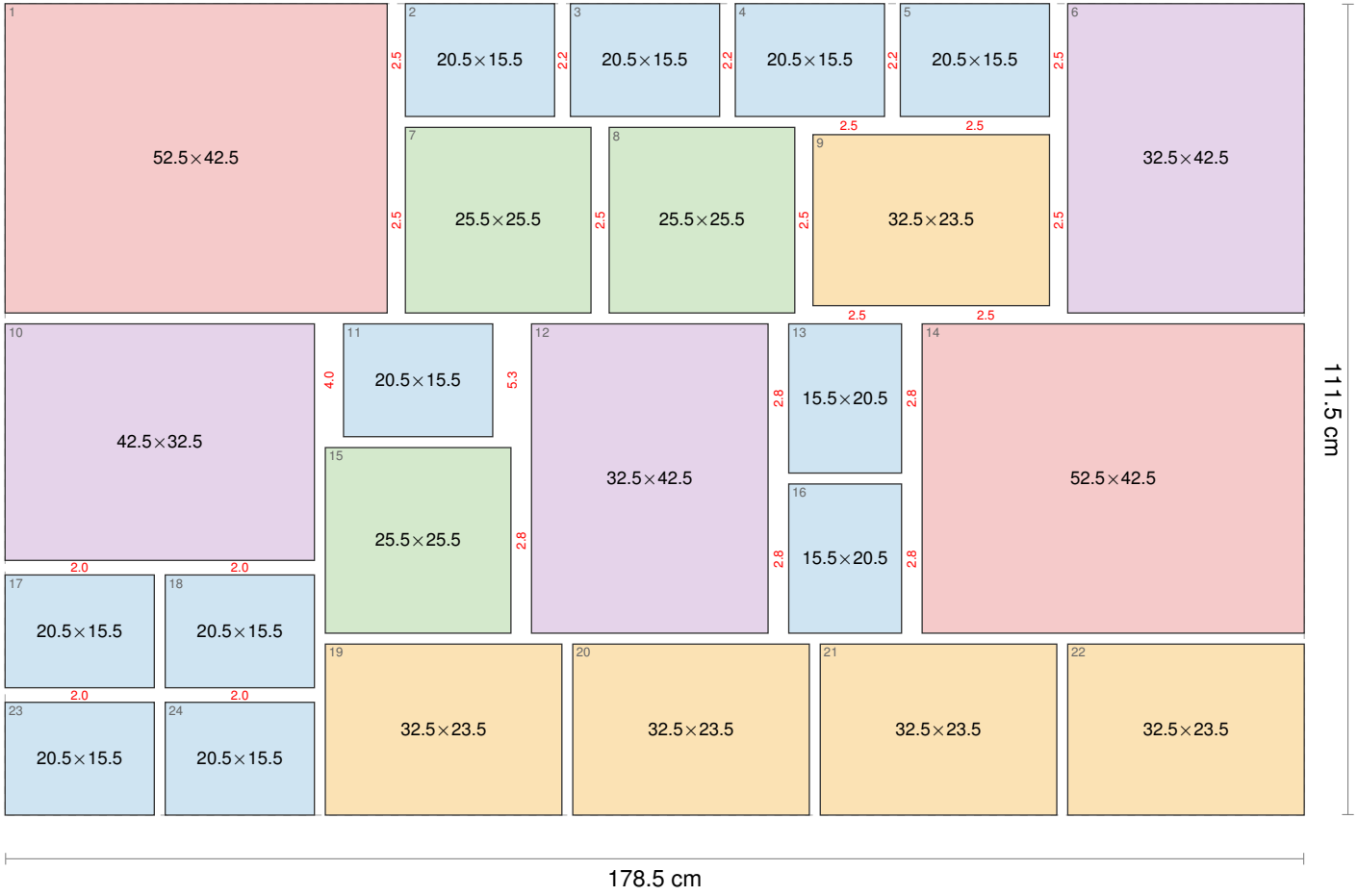
Outer size	169.0 × 116.5 cm
Nominal gap	2 cm
Actual gaps	2.0 – 4.5 cm
Max. deviation	2.5 cm
Mean deviation	0.69 cm
Border notches	left 8 cm
Corners filled	4 / 4
Frames used	24 of 28
Large frames	2, not touching, min. dist. 116 cm
Entropy	0.76
mix / variety	0.76 / 0.73
stagger / lines	0.56 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	2	2
	32.5×42.5	P	1	2
	42.5×32.5	L	1	1
	32.5×23.5	L	4	5
	23.5×32.5	P	0	1
	25.5×25.5		5	5
	42.5×52.5	P	0	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	20.5 × 15.5	0.0	0.0
2	20.5 × 15.5	22.5	0.0
3	25.5 × 25.5	45.2	0.0
4	32.5 × 23.5	73.2	0.0
5	25.5 × 25.5	108.2	0.0
6	15.5 × 20.5	136.0	0.0
7	15.5 × 20.5	153.5	0.0
8	20.5 × 15.5	0.0	19.5
9	20.5 × 15.5	22.5	19.5
10	32.5 × 23.5	136.5	22.5
11	25.5 × 25.5	80.2	28.0
12	25.5 × 25.5	108.2	28.0
13	32.5 × 23.5	45.2	29.0
14	42.5 × 32.5	0.0	39.0
15	32.5 × 23.5	136.5	48.0
16	20.5 × 15.5	45.2	56.0
17	20.5 × 15.5	67.9	56.0
18	20.5 × 15.5	90.6	56.0
19	20.5 × 15.5	113.2	56.0
20	20.5 × 15.5	91.5	73.5
21	52.5 × 42.5	0.0	74.0
22	32.5 × 42.5	54.5	74.0
23	52.5 × 42.5	116.5	74.0
24	25.5 × 25.5	89.0	91.0

Layout 12 178.5 × 111.5 cm gap 1.5 cm 24 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

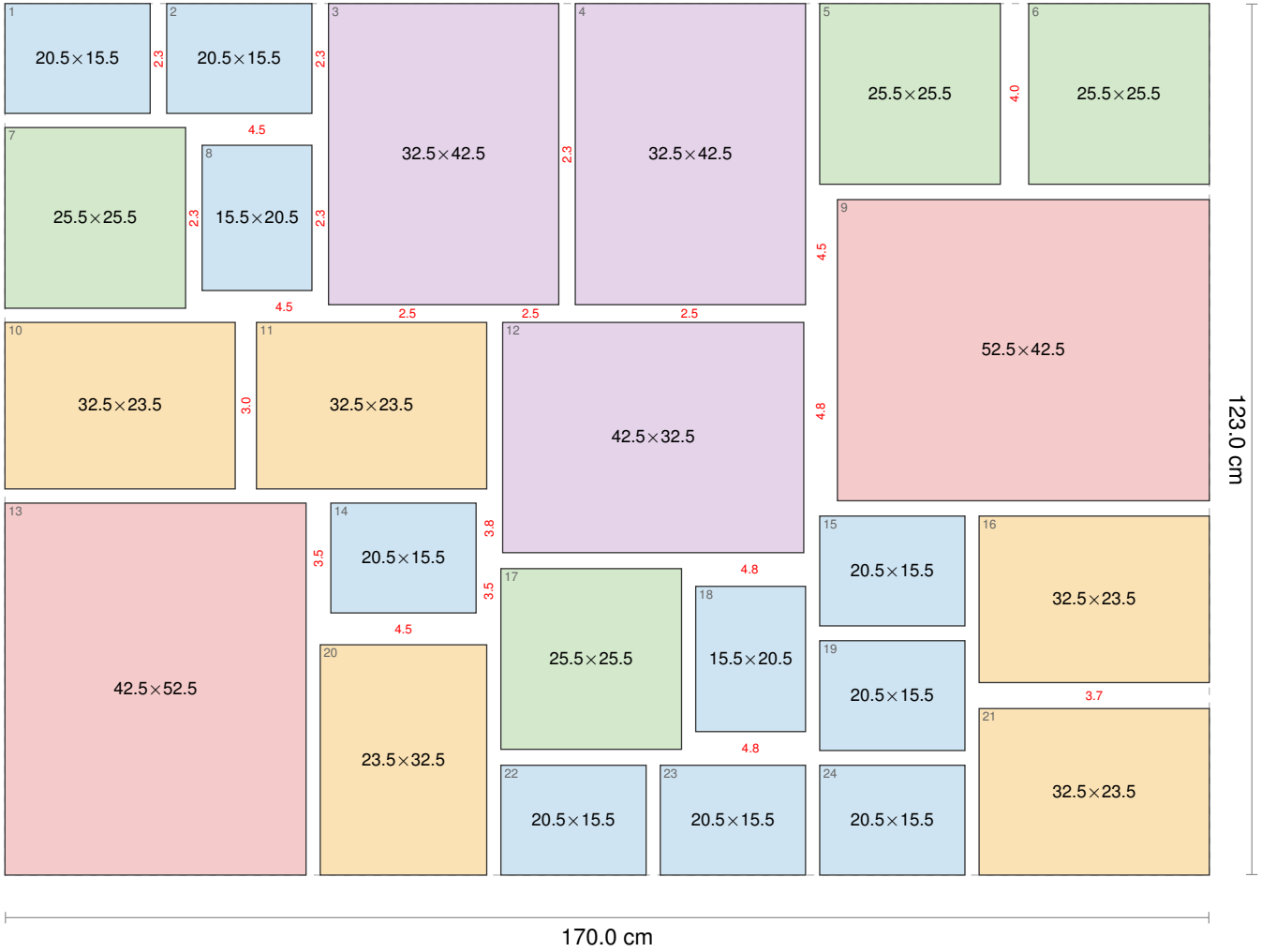
Outer size	178.5 × 111.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 5.3 cm
Max. deviation	3.8 cm
Mean deviation	0.55 cm
Border notches	none
Corners filled	4 / 4
Frames used	24 of 28
Large frames	2, not touching, min. dist. 133 cm
Entropy	0.76
mix / variety	0.78 / 0.71
stagger / lines	0.53 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	9	9
	15.5×20.5	P	2	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	0	1
	25.5×25.5		3	5
	42.5×52.5	P	0	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	52.5 × 42.5	0.0	0.0
2	20.5 × 15.5	55.0	0.0
3	20.5 × 15.5	77.7	0.0
4	20.5 × 15.5	100.3	0.0
5	20.5 × 15.5	123.0	0.0
6	32.5 × 42.5	146.0	0.0
7	25.5 × 25.5	55.0	17.0
8	25.5 × 25.5	83.0	17.0
9	32.5 × 23.5	111.0	18.0
10	42.5 × 32.5	0.0	44.0
11	20.5 × 15.5	46.5	44.0
12	32.5 × 42.5	72.3	44.0
13	15.5 × 20.5	107.7	44.0
14	52.5 × 42.5	126.0	44.0
15	25.5 × 25.5	44.0	61.0
16	15.5 × 20.5	107.7	66.0
17	20.5 × 15.5	0.0	78.5
18	20.5 × 15.5	22.0	78.5
19	32.5 × 23.5	44.0	88.0
20	32.5 × 23.5	78.0	88.0
21	32.5 × 23.5	112.0	88.0
22	32.5 × 23.5	146.0	88.0
23	20.5 × 15.5	0.0	96.0
24	20.5 × 15.5	22.0	96.0

Layout 13 170.0 × 123.0 cm gap 2 cm 24 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

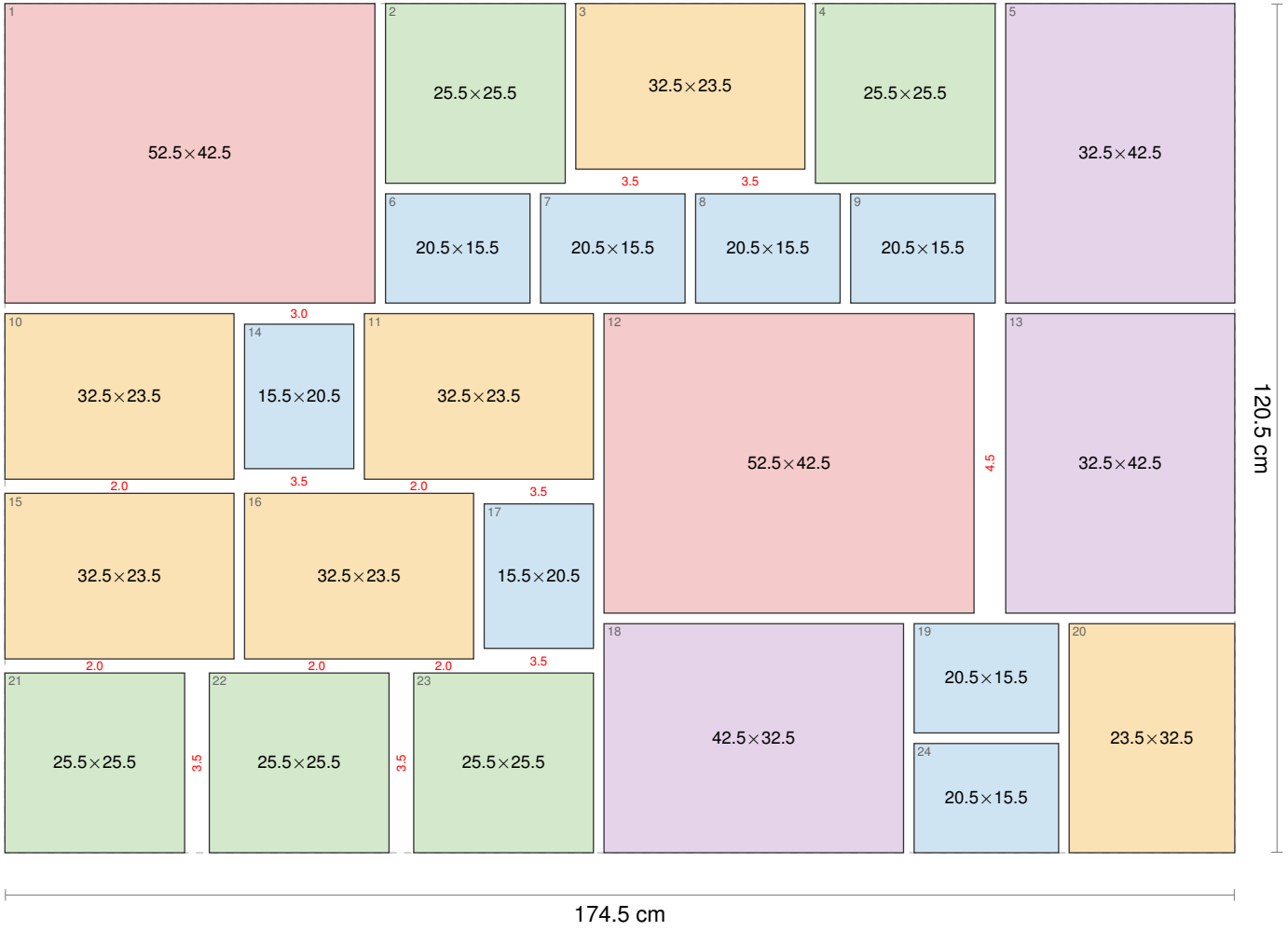
Outer size	170.0 × 123.0 cm
Nominal gap	2 cm
Actual gaps	2.0 – 4.8 cm
Max. deviation	2.8 cm
Mean deviation	0.67 cm
Border notches	top 4 cm, right 4 cm
Corners filled	4 / 4
Frames used	24 of 28
Large frames	2, not touching, min. dist. 132 cm
Entropy	0.82
mix / variety	0.87 / 0.81
stagger / lines	0.61 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	8	9
	15.5×20.5	P	2	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	4	5
	23.5×32.5	P	1	1
	25.5×25.5		4	5
	42.5×52.5	P	1	1
	52.5×42.5	L	1	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	20.5 × 15.5	0.0	0.0
2	20.5 × 15.5	22.8	0.0
3	32.5 × 42.5	45.7	0.0
4	32.5 × 42.5	80.5	0.0
5	25.5 × 25.5	115.0	0.0
6	25.5 × 25.5	144.5	0.0
7	25.5 × 25.5	0.0	17.5
8	15.5 × 20.5	27.8	20.0
9	52.5 × 42.5	117.5	27.7
10	32.5 × 23.5	0.0	45.0
11	32.5 × 23.5	35.5	45.0
12	42.5 × 32.5	70.2	45.0
13	42.5 × 52.5	0.0	70.5
14	20.5 × 15.5	46.0	70.5
15	20.5 × 15.5	115.0	72.3
16	32.5 × 23.5	137.5	72.3
17	25.5 × 25.5	70.0	79.8
18	15.5 × 20.5	97.5	82.2
19	20.5 × 15.5	115.0	89.9
20	23.5 × 32.5	44.5	90.5
21	32.5 × 23.5	137.5	99.5
22	20.5 × 15.5	70.0	107.5
23	20.5 × 15.5	92.5	107.5
24	20.5 × 15.5	115.0	107.5

Layout 14 174.5 × 120.5 cm gap 1.5 cm 24 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

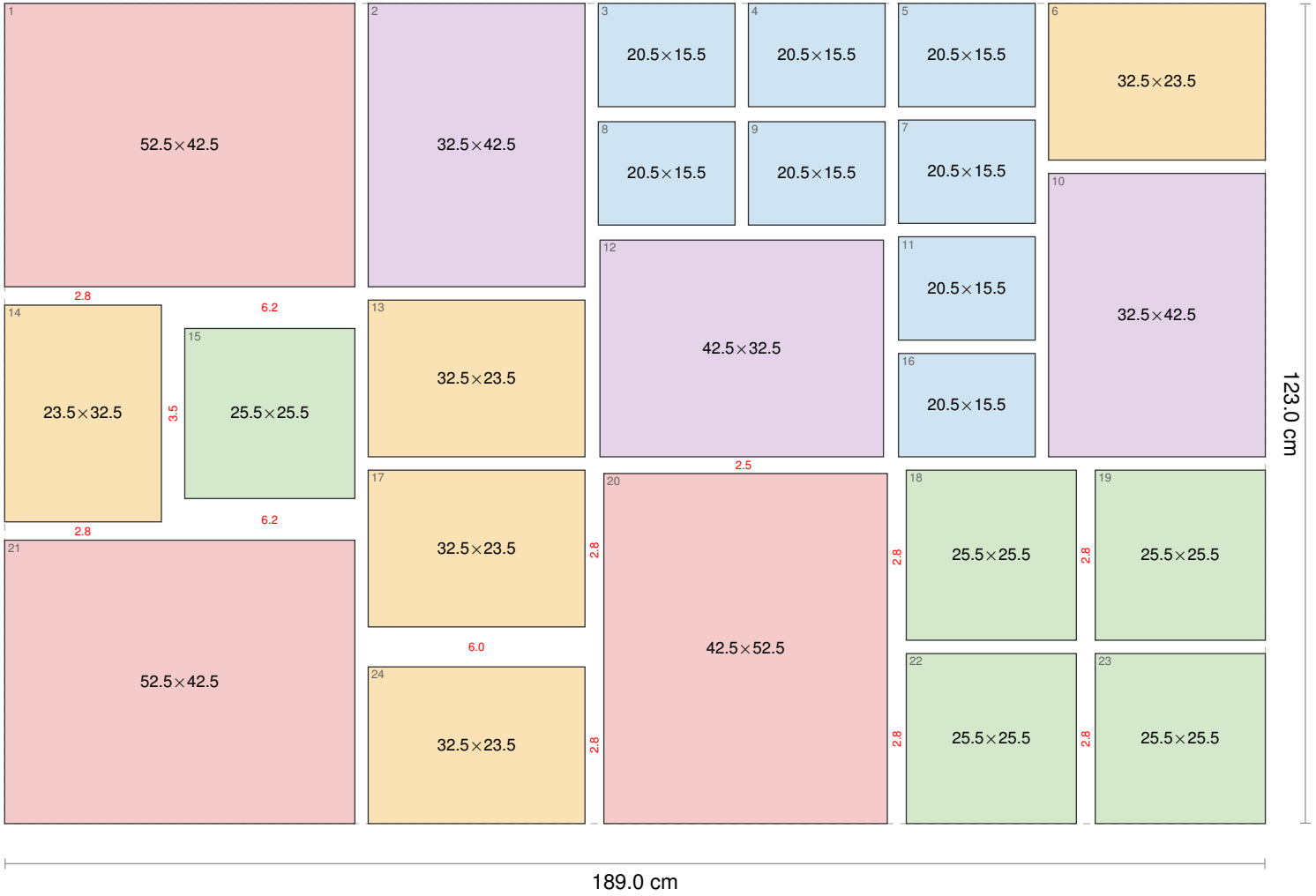
Outer size	174.5 × 120.5 cm
Nominal gap	1.5 cm
Actual gaps	1.5 – 4.5 cm
Max. deviation	3.0 cm
Mean deviation	0.44 cm
Border notches	bottom 7 cm
Corners filled	4 / 4
Frames used	24 of 28
Large frames	2, not touching, min. dist. 96 cm
Entropy	0.80
mix / variety	0.86 / 0.79
stagger / lines	0.54 / 1.00 (1 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	6	9
	15.5×20.5	P	2	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	5	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	0	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	52.5 × 42.5	0.0	0.0
2	25.5 × 25.5	54.0	0.0
3	32.5 × 23.5	81.0	0.0
4	25.5 × 25.5	115.0	0.0
5	32.5 × 42.5	142.0	0.0
6	20.5 × 15.5	54.0	27.0
7	20.5 × 15.5	76.0	27.0
8	20.5 × 15.5	98.0	27.0
9	20.5 × 15.5	120.0	27.0
10	32.5 × 23.5	0.0	44.0
11	32.5 × 23.5	51.0	44.0
12	52.5 × 42.5	85.0	44.0
13	32.5 × 42.5	142.0	44.0
14	15.5 × 20.5	34.0	45.5
15	32.5 × 23.5	0.0	69.5
16	32.5 × 23.5	34.0	69.5
17	15.5 × 20.5	68.0	71.0
18	42.5 × 32.5	85.0	88.0
19	20.5 × 15.5	129.0	88.0
20	23.5 × 32.5	151.0	88.0
21	25.5 × 25.5	0.0	95.0
22	25.5 × 25.5	29.0	95.0
23	25.5 × 25.5	58.0	95.0
24	20.5 × 15.5	129.0	105.0

Layout 15 189.0 × 123.0 cm gap 2 cm 24 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

Outer size	189.0 × 123.0 cm
Nominal gap	2 cm
Actual gaps	2.0 – 6.2 cm
Max. deviation	4.2 cm
Mean deviation	0.50 cm
Border notches	none
Corners filled	4 / 4
Frames used	24 of 28
Large frames	3, not touching, min. dist. 80 cm
Entropy	0.62
mix / variety	0.82 / 0.67
stagger / lines	0.50 / 0.50 (3 through)

	outer w × h (cm)		used	have
	20.5×15.5	L	8	9
	15.5×20.5	P	0	2
	32.5×42.5	P	2	2
	42.5×32.5	L	1	1
	32.5×23.5	L	4	5
	23.5×32.5	P	1	1
	25.5×25.5		5	5
	42.5×52.5	P	1	1
	52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	52.5 × 42.5	0.0	0.0
2	32.5 × 42.5	54.5	0.0
3	20.5 × 15.5	89.0	0.0
4	20.5 × 15.5	111.5	0.0
5	20.5 × 15.5	134.0	0.0
6	32.5 × 23.5	156.5	0.0
7	20.5 × 15.5	134.0	17.5
8	20.5 × 15.5	89.0	17.8
9	20.5 × 15.5	111.5	17.8
10	32.5 × 42.5	156.5	25.5
11	20.5 × 15.5	134.0	35.0
12	42.5 × 32.5	89.2	35.5
13	32.5 × 23.5	54.5	44.5
14	23.5 × 32.5	0.0	45.2
15	25.5 × 25.5	27.0	48.8
16	20.5 × 15.5	134.0	52.5
17	32.5 × 23.5	54.5	70.0
18	25.5 × 25.5	135.2	70.0
19	25.5 × 25.5	163.5	70.0
20	42.5 × 52.5	89.8	70.5
21	52.5 × 42.5	0.0	80.5
22	25.5 × 25.5	135.2	97.5
23	25.5 × 25.5	163.5	97.5
24	32.5 × 23.5	54.5	99.5

Layout 16 170.5 × 126.5 cm gap 2.5 cm 23 frames



Scale 1:10 (1 mm = 1 cm). Red numbers mark gaps that deviate from the nominal gap by more than 0.25 cm. Small grey numbers index the coordinate table.

Outer size	170.5 × 126.5 cm
Nominal gap	2.5 cm
Actual gaps	2.5 – 6.0 cm
Max. deviation	3.5 cm
Mean deviation	0.77 cm
Border notches	none
Corners filled	4 / 4
Frames used	23 of 28
Large frames	3, not touching, min. dist. 79 cm
Entropy	0.72
mix / variety	0.79 / 0.80
stagger / lines	0.55 / 0.75 (2 through)

outer w × h (cm)		used	have
20.5×15.5	L	9	9
15.5×20.5	P	1	2
32.5×42.5	P	1	2
42.5×32.5	L	1	1
32.5×23.5	L	4	5
23.5×32.5	P	0	1
25.5×25.5		4	5
42.5×52.5	P	1	1
52.5×42.5	L	2	2

Positions (top-left corner, cm)

#	frame w × h	x	y
1	52.5 × 42.5	0.0	0.0
2	25.5 × 25.5	55.0	0.0
3	32.5 × 42.5	83.0	0.0
4	52.5 × 42.5	118.0	0.0
5	20.5 × 15.5	57.5	28.0
6	25.5 × 25.5	0.0	46.0
7	25.5 × 25.5	64.0	46.0
8	32.5 × 23.5	92.0	46.0
9	25.5 × 25.5	145.0	46.0
10	32.5 × 23.5	28.5	47.0
11	15.5 × 20.5	127.0	48.5
12	42.5 × 52.5	128.0	74.0
13	20.5 × 15.5	0.0	74.5
14	42.5 × 32.5	23.5	74.5
15	20.5 × 15.5	69.0	74.5
16	32.5 × 23.5	92.0	74.5
17	20.5 × 15.5	0.0	92.8
18	20.5 × 15.5	69.0	92.8
19	32.5 × 23.5	92.0	103.0
20	20.5 × 15.5	0.0	111.0
21	20.5 × 15.5	23.0	111.0
22	20.5 × 15.5	46.0	111.0
23	20.5 × 15.5	69.0	111.0